



## DPUSGEDistortionBezier5

Distorts the image of a SGEView using two-dimensional Bezier interpolation.

### 1. Dynamic parameters:

*X[Col][Row],  
Y[Col][Row]*

Coordinates (in the 2D overlay coordinate system), where the control point is placed. In total, 5x5 control points are used to parameterize the Bezier curves. Therefore, *Col* and *Row* both run from 0 up to 4.

*NDX, NDY*

Coordinates (in the 2D overlay coordinate system) where the top-left edge of the undistorted scenery is placed, if an additional undistorted output is desired. (See below)

*NDW, NDH  
TexSMin, TexSMax*

Width and height (in pixels) of the additional undistorted output. Specifies which range of the computed image (in horizontal dimension) is actually displayed.

*TexTMin, TexTMax*

Default: (0...1), i.e. the whole computed image is displayed. Specifies which range of the computed image (in vertical dimension) is actually displayed. Default: (0...1), i.e. the whole computed image is displayed.

### 2. Static parameters:

*ViewID*

The number of the view whose image is distorted. Counting starts at 0, which is assigned to the first view that is listed in the configuration.

*ResolutionX, ResolutionY*

Size of the off-screen image buffer (in pixels), into which the scenery is rendered. Should correspond to the image size after distortion. Large Resolution values reduce the achievable frame rate.

Note: For older graphics boards (e.g. NVidia GeForce 6000, 7000 series), powers of two must be used for ResolutionX and ResolutionY, otherwise no output can be seen. Newer boards can use arbitrary values.

*SubdivX, SubdivY*

Subdivision of the projection geometry in X and Y direction. The more the image is distorted, the more subdivisions are required; however, that reduces the achievable frame rate.

*DistortResetViewport*

The default value is 50 and is sufficient for most practical purposes. Selects whether the coordinates *Xnn*, *Ynn* and *NDX* ... *NDH* are relative to the whole window into which the view is rendered (1) or only to the selected viewport (0).

The default value is 0 and only needs to be changed if an additional undistorted output is desired. (See below)



### 3. Dependencies to other DPUs:

**DPUSGEDistortionBezier5** must run together with **DPUSGEWorld** and **DPUSGEView**.

The following Index values must be assigned:

**DPUSGEWorld.Index < DPUSGEDistortionBezier5.Index < DPUSGEView.Index**

### 4. Example:

The DPU works well with the following script:

```
DPUSGEDistortionBezier5 SGEDistortion
{
    Computer = {VIS};
    Index = 50;

    ResolutionX = 1024;
    ResolutionY = 1024;

    X00 = 0; X10 = 220; X20 = 440; X30 = 660; X40 = 880;
    Y00 = 0; Y10 = 10; Y20 = 100; Y30 = 10; Y40 = 0;

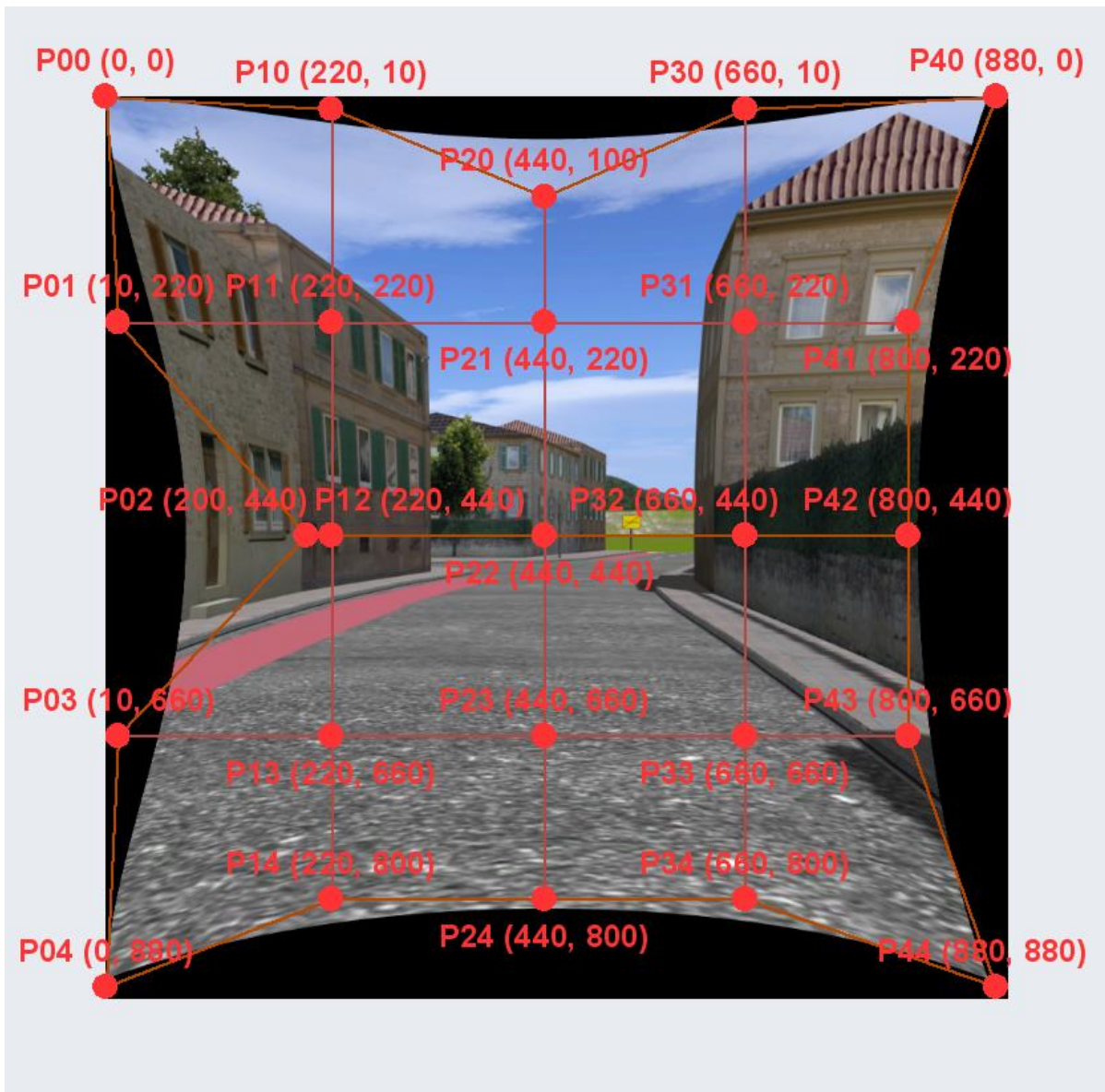
    X01 = 10; X11 = 220; X21 = 440; X31 = 600; X41 = 800;
    Y01 = 220; Y11 = 220; Y21 = 220; Y31 = 220; Y41 = 220;

    X02 = 200; X12 = 220; X22 = 440; X32 = 600; X42 = 800;
    Y02 = 440; Y12 = 440; Y22 = 440; Y32 = 440; Y42 = 440;

    X03 = 10; X13 = 220; X23 = 440; X33 = 600; X43 = 800;
    Y03 = 660; Y13 = 660; Y23 = 660; Y33 = 660; Y43 = 660;

    X04 = 0; X14 = 220; X24 = 440; X34 = 660; X44 = 880;
    Y04 = 880; Y14 = 800; Y24 = 800; Y34 = 800; Y44 = 880;
};
```

Assuming that the associated SGEView has a resolution of 900 x 900 pixels, the scenery is slightly compressed using these parameters („pincushion distortion”). The result is visible in the following screenshot.



Please note that only the edge points (P00, P40, P04, and P44) are located exactly on the border of the displayed area. The other control points influence the Bezier curves more indirectly.

## 5. Additional undistorted output:

In many cases, it is useful to display not only the distorted scenery (on a projector), but additionally the undistorted scenery (on a computer monitor that has been connected in parallel).

DPUSGEDistortionBezier5 supports this configuration without a noticeable performance loss.

The necessary system and configuration settings for the undistorted output are described in the document for DPUSGEDistortionQuad. The same method also works for DPUSGEDistortionBezier5.